Basic Discrete-Time Fourier Transform Pairs

Time Domain $x[n] = \frac{1}{2\pi}\int_{<2\pi>} X(e^{j\omega})e^{j\omega n}d\omega$	Frequency Domain $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$
$x[n] = \begin{cases} 1, & \lvert n\rvert \le M \\ 0, & \text{otherwise} \end{cases}$	$X(e^{j\omega}) = \frac{\sin[\omega(\frac{2M+1}{2})]}{\sin(\frac{\omega}{2})}$
$x[n] = \alpha^n u[n], \quad \lvert\alpha\rvert < 1$	$X(e^{j\omega}) = \frac{1}{1-\alpha e^{-j\omega}}$
$x[n] = \delta[n]$	$X(e^{j\omega}) = 1$
$x[n] = u[n]$	$X(e^{j\omega}) = \frac{1}{1-e^{-j\omega}} + \pi\sum_{p=-\infty}^{\infty}\delta(\omega - 2\pi p)$
$x[n] = \frac{1}{\pi n}\sin(Wn), \quad 0 < W \le \pi$	$X(e^{j\omega}) = \begin{cases} 1, & \lvert\omega\rvert < W \\ 0, & W < \lvert\omega\rvert \le \pi \end{cases}$ $X(e^{j\omega})$ is 2π periodic
$x[n] = (n+1)\alpha^n u[n]$	$X(e^{j\omega}) = \frac{1}{(1-\alpha e^{-j\omega})^2}$

Discrete-Time Fourier Transform Pairs for Periodic Signals

Periodic Time Domain Signal	Discrete-Time Fourier Transform
$x[n] = \sum_{k=<N>} X[k] e^{jk\omega_o n}$	$X(e^{j\omega}) = 2\pi \sum_{k=-\infty}^{\infty} X[k]\delta(\omega - k\omega_o)$
$x[n] = \cos(\omega_1 n)$	$X(e^{j\omega}) = \pi \sum_{k=-\infty}^{\infty} \delta(\omega - \omega_1 - k2\pi)$ $+\delta(\omega + \omega_1 - k2\pi)$
$x[n] = \sin(\omega_1 n)$	$X(e^{j\omega}) = \frac{\pi}{j} \sum_{k=-\infty}^{\infty} \delta(\omega - \omega_1 - k2\pi)$ $-\delta(\omega + \omega_1 - k2\pi)$
$x[n] = e^{j\omega_1 n}$	$X(e^{j\omega}) = 2\pi \sum_{k=-\infty}^{\infty} \delta(\omega - \omega_1 - k2\pi)$
$x[n] = \sum_{k=-\infty}^{\infty} \delta[n - kN]$	$X(e^{j\omega}) = \frac{2\pi}{N} \sum_{k=-\infty}^{\infty} \delta(\omega - \frac{k2\pi}{N})$